

SHERLOCKML / AUTONOMOUS ML INCIDENT COMMAND

Demo Walkthrough

An autonomous AI detective and doctor for machine-learning systems — a screen-by-screen guide to the live investigation: evidence, debate, diagnosis, treatment, and validated recovery.

CASE-DRIFT-001 · LIVE RUN

LANGGRAPH MULTI-AGENT

FASTAPI + STREAMLIT

HUMAN APPROVAL REQUIRED

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SherlockML stages a realistic ML reliability incident against a synthetic fraud-detection model, then lets a team of specialist agents work the case: a **Detective** gathers evidence, a **Statistician** measures the data, an **Infra agent** rules out the platform, a **War Room** debates the root cause, an **Engineer** writes a bounded repair, an **Experiment agent** proves the fix on real retrained models, and a **Model Doctor** signs the treatment plan — always ending in *approval for human review*, never an automatic deployment.

Everything below was captured from a **live run** of the engine (`CASE-DRIFT-001` , the data-drift scenario), not from mockups.

1. Launch the command room

```
python3 -m venv .venv && source .venv/bin/activate
make install          # installs the package plus dev tooling
make run-api          # terminal 1 - FastAPI engine on http://127.0.0.1:8788
make run-ui           # terminal 2 - Streamlit control room on http://127.0.0.1:8502
```

Open `http://127.0.0.1:8502`. The dashboard boots into a clearly labelled **interactive preview** (`PREVIEW-001`) so the room is never empty; the moment you run a real investigation, every panel switches to live engine output and the hero banner flips to `LIVE INVESTIGATION`.

To start the demo: in the sidebar under *Scenario control*, keep **Data drift** selected and press 🔍 **Investigate incident**. A status panel advances through each agent — evidence collection, statistical analysis, the debate, a pipeline repair, retraining, validation, and the final report — in a few seconds on a laptop.

Optional: tick **Demo failed-recovery path (strict gates)** to rehearse a candidate that fails the F1 improvement gate and routes through the escalation hold before the doctor signs off.

2. Model Health — the patient's bedside monitor

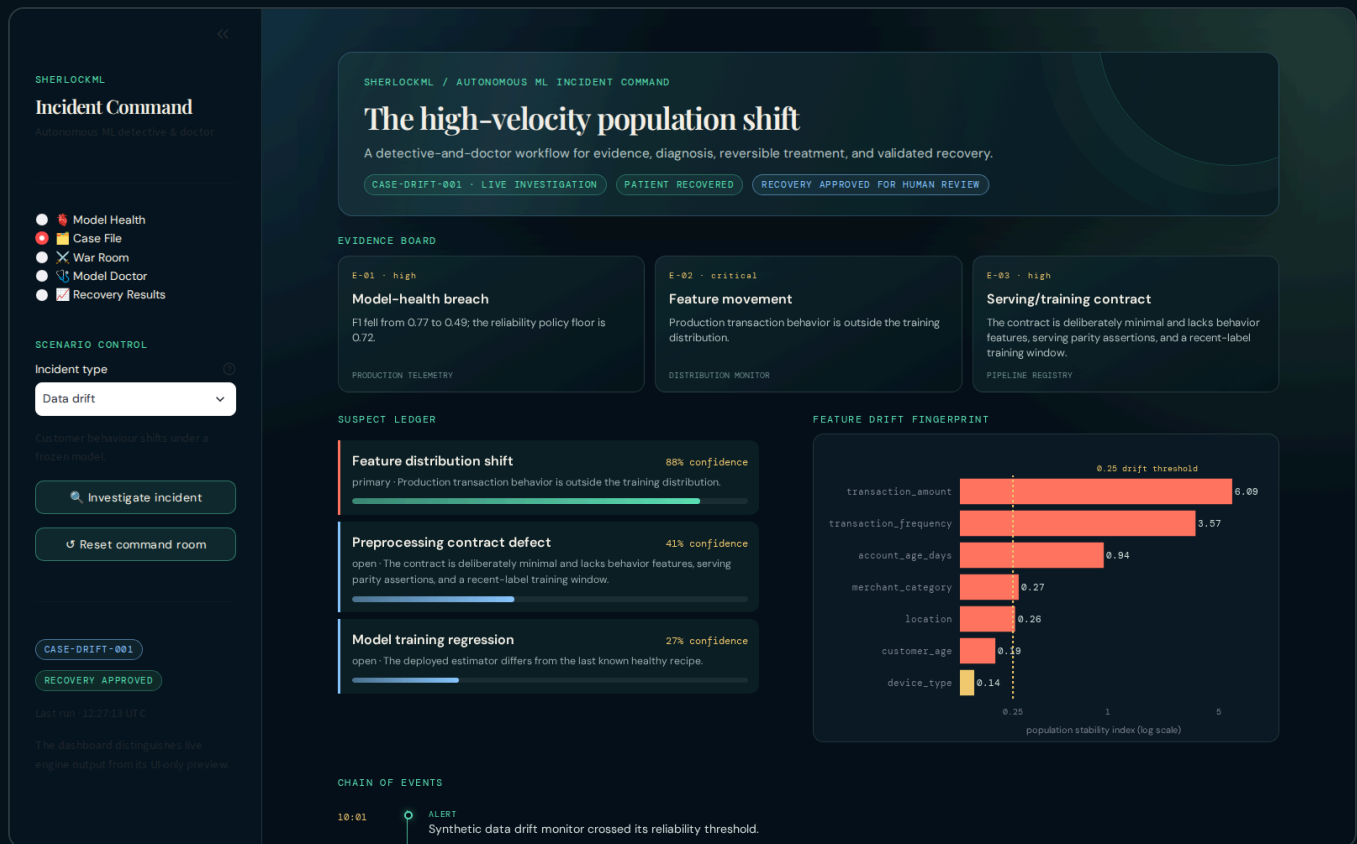


The first page frames the incident as a medical case.

- **Hero banner.** The incident title generated for this case (“The high-velocity population shift”), the case id chip (CASE-DRIFT-001 · LIVE INVESTIGATION), the patient state (PATIENT RECOVERED), and the workflow status (RECOVERY APPROVED FOR HUMAN REVIEW).
- **Phase stepper.** Six stages — ALERT → EVIDENCE → DEBATE → REPAIR → VERIFY → REPORT — derived from the run’s timeline. Completed phases glow mint; the current phase is gold.
- **Patient avatar.** The model itself, with a pulsing status ring and an animated ECG line. It is coral and feverish (🤒) while the case is critical and settles to mint (😌) once the candidate clears its gates.
- **Health gauge.** The model’s health score (F1 expressed as a percentage) on a dial with red/amber/green risk bands. The gold needle marks the pre-incident baseline, and the delta below the number shows the recovery relative to that baseline.
- **Vital signs.** Four cards: the pre-incident baseline (77%), health at the moment of the alert (49%, -29 points), the recovery signal after treatment (78%, +30 points of candidate lift), and serving latency with the workflow status.
- **Investigation pulse.** The chronological event feed. Each entry is one agent completing its step — the alert, the statistician’s ranking, the infra rule-out, the debate, the repair, the verification, the decision, the record, and the report.

What to say: “The model hasn’t crashed — it still answers every request. Its F1 quietly fell from 77% to 49%. That silent failure is exactly what SherlockML is built to catch.”

3. Case File — evidence before conclusions



The Detective's output: observable evidence first, hypotheses second.

- **Evidence board.** Three cited exhibits, each with a source and strength:
- **E-01 — Model-health breach** (production telemetry): F1 fell from 0.77 to 0.49 against a 0.72 policy floor.
- **E-02 — Feature movement** (distribution monitor): production behaviour is outside the training distribution.
- **E-03 — Serving/training contract** (pipeline registry): the versioned feature contract is minimal and unguarded.
- **Suspect ledger.** The three competing root-cause hypotheses with animated confidence bars. Scores are **computed from evidence** — PSI severity, missing-rate signals, contract gaps, and model metadata — not looked up from a table. The leading suspect is highlighted in coral; ruled-out hypotheses stay on the board with their rationale.
- **Feature drift fingerprint.** The Statistician's real measurements: one PSI (population stability index) bar per feature, colour-coded by severity, on a log scale with the industry 0.25 material-drift threshold drawn as a dotted line. In this scenario `transaction_amount` reads PSI 6.09 — roughly 24× beyond the threshold — because the synthetic population genuinely shifted. These numbers come from actual PSI/KS computations over the generated dataframes, not hand-typed values.
- **Chain of events.** A compact copy of the timeline for cross-reference.

What to say: *"The detective doesn't guess. Three exhibits, three suspects, each with a confidence score — and the statistician's chart shows exactly which features moved and by how much."*

4. War Room — recorded disagreement, then consensus

The screenshot displays the SHERLOCKML War Room interface. On the left, a sidebar titled 'Incident Command' shows navigation options: Model Health, Case File, War Room (selected), Model Doctor, and Recovery Results. Below this is 'SCENARIO CONTROL' with an 'Incident type' dropdown set to 'Data drift' and buttons for 'Investigate incident' and 'Reset command room'. The main area is titled 'The high-velocity population shift' and shows a transcript of agent interactions. The transcript includes: Sherlock (Detective Agent) stating the highest-confidence suspect is feature distribution shift (88%); Ada (Statistician Agent) providing statistical signals (PSI=609 and KS p=0); Linus (ML Engineer Agent) dissenting, wanting to rule out preprocessing contract defect; SRE-7 (Infra Agent) reporting infrastructure is within its operating envelope; and Moriarty (War Room Moderator) reaching consensus and authorizing a reversible repair. At the bottom, a 'MODERATOR CONCLUSION' card shows an 88% confidence ring and states the primary suspect is feature distribution shift, with production transaction behavior outside the training distribution.

SHERLOCKML / AUTONOMOUS ML INCIDENT COMMAND

The high-velocity population shift

A detective-and-doctor workflow for evidence, diagnosis, reversible treatment, and validated recovery.

CASE-DRIFT-001 · LIVE INVESTIGATION · PATIENT RECOVERED · RECOVERY APPROVED FOR HUMAN REVIEW

LIVE MULTI-AGENT TRANSCRIPT

Sherlock Detective Agent OPENS CASE
The highest-confidence suspect is feature distribution shift (88%).

Ada Statistician Agent EVIDENCE
transaction_amount is the strongest statistical signal: PSI=609 and KS p=0.

Linus ML Engineer Agent DISSENT
I want to rule out preprocessing contract defect before touching production. I will inspect the versioned feature contract and test the smallest safe repair.

SRE-7 Infra Agent OPERATIONAL CHECK
Serving infrastructure is within its operating envelope; no API or resource event explains the model-quality loss.

Moriarty War Room Moderator CONSENSUS
Consensus reached: Feature distribution shift is the primary cause for incident type 'data_drift'. Proceed with a reversible repair and validation gate.

MODERATOR CONCLUSION

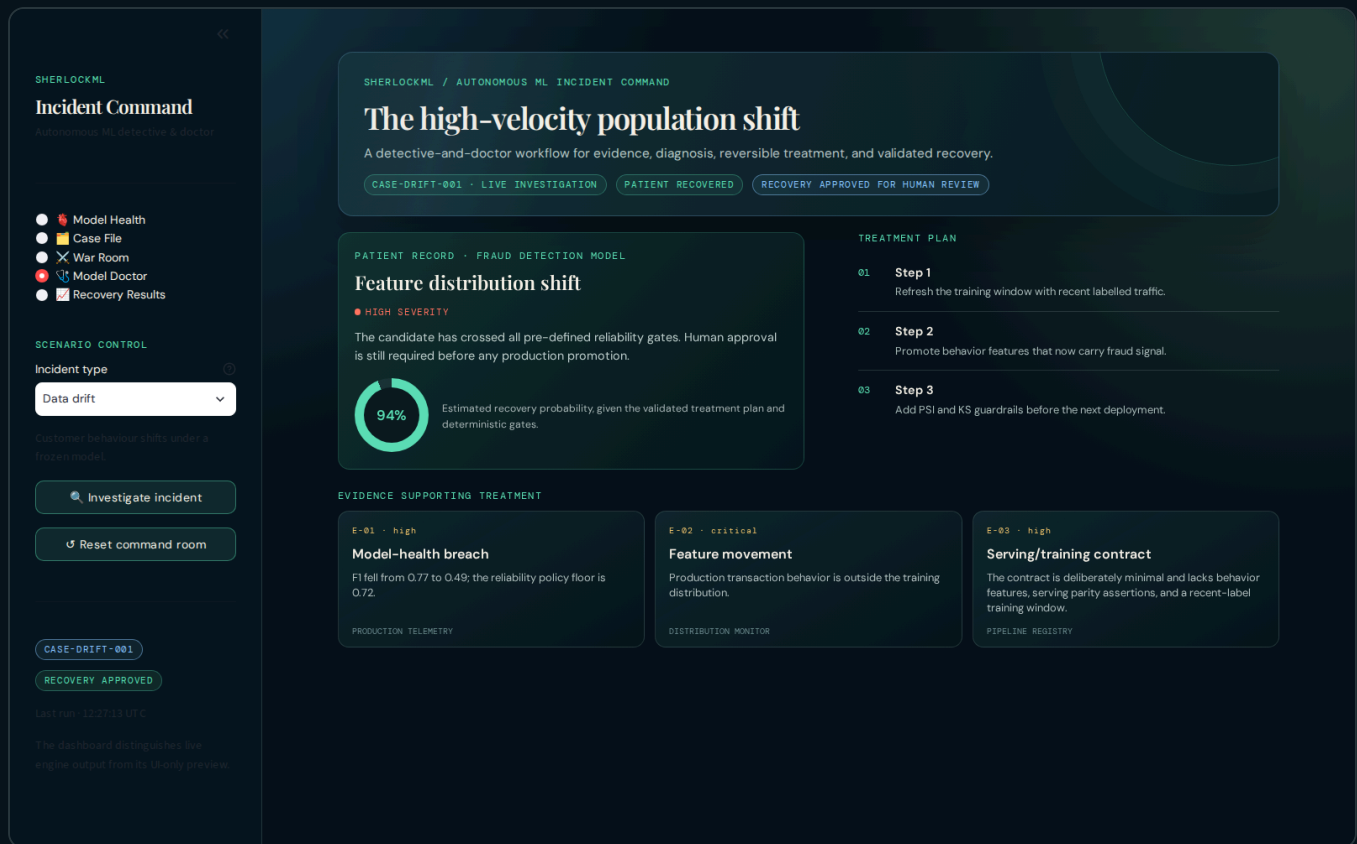
WAR ROOM VERDICT
Feature distribution shift
Production transaction behavior is outside the training distribution.

Root-cause analysis as a transcript rather than a single opaque answer.

- **The transcript.** Six agents speak in turn, each in a chat bubble with an avatar and a stance chip:
- **Sherlock** (Detective, OPENS CASE) states the leading suspect and its **computed** confidence.
- **Ada** (Statistician, EVIDENCE) brings the measurements: PSI and the KS test verdict for the strongest feature (categorical features are labelled KS N/A).
- **Linus** (ML Engineer, DISSENT, coral chip) pushes back — he wants the pipeline contract ruled out before anyone touches production.
- **Linus** again (CONTRACT CHECK) cross-examines the versioned feature contract against the active preprocessor and champion schema.
- **SRE-7** (Infra, OPERATIONAL CHECK) reports latency, error rate, and memory are inside the operating envelope — clearing the platform as a suspect.
- **Moriarty** (Moderator, CONSENSUS, mint chip) closes the case only after the claims are compared, cites the confidence margin between the top two suspects, and authorises a *reversible* repair with validation gates.
- **Moderator conclusion.** The verdict card: the primary suspect with a confidence ring and the one-line rationale.

What to say: “Notice the red chip — the engineer disagrees, on the record. The moderator only resolves the case after the dissent is answered. That’s how real incident reviews work.”

5. Model Doctor — diagnosis and prescription



The technical verdict translated into a treatment plan.

- **Patient record.** The diagnosed condition (*Feature distribution shift*), its severity (HIGH), the bedside note, and a **recovery-probability ring** estimated from gate pass ratio and candidate lift — not a fixed placeholder.
- **Treatment plan.** Three numbered prescriptions for this diagnosis: refresh the training window with recent labelled traffic, promote the behaviour features that now carry the fraud signal, and install PSI/KS guardrails before the next deployment.
- **Evidence supporting treatment.** The exhibits that justify the prescription, carried over from the case file so the plan is auditable.

What to say: *“Like a real doctor, it doesn’t just name the disease — it prescribes a treatment, estimates the odds, and cites the evidence for both.”*

6. Recovery Results — proof, not promises



The Experiment agent actually retrains a candidate model and evaluates it on the same deterministic incident window. This page is the receipt.

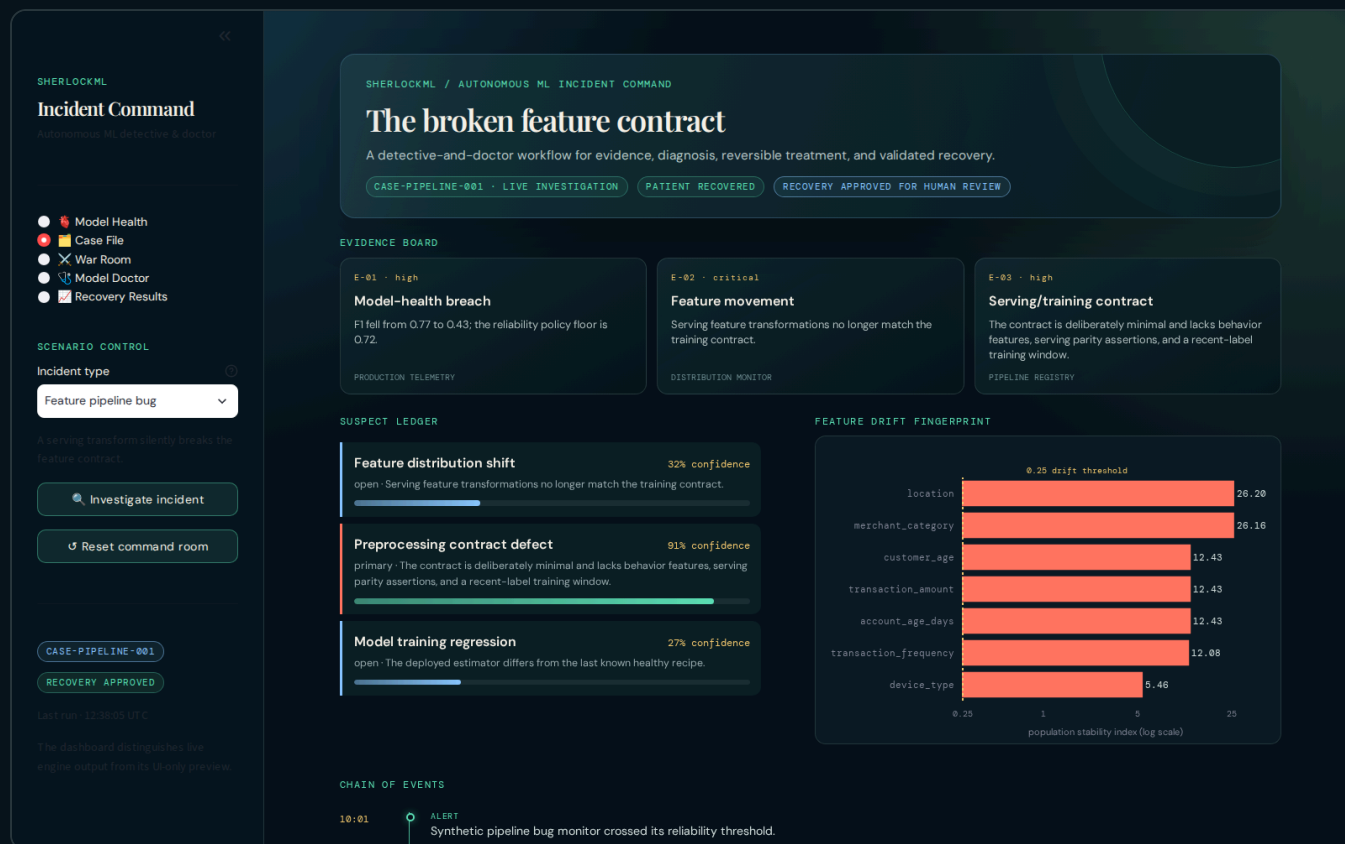
- **Baseline vs candidate.** Grouped bars for accuracy, precision, recall and F1: the failed **v17-production** model (coral) against the treated **v18-candidate** (mint). Below, four delta chips summarise the movement — in this run F1 +29.9 points and precision +49.1 points.
- A subtle but honest detail: the **failed** baseline can show very high recall — a model that flags almost everything catches every fraud while being useless. Precision is what collapsed, and the chart makes that visible.
- **Health trajectory.** The three-act arc as a chart: 77% → 49% → 78%.
- **Validation gates.** A pass/fail board for the explicit recovery gates: minimum F1 improvement, F1 floor, and recall floor. Each row shows the measured value against the threshold.
- **Approval banner.** **RECOVERY APPROVED FOR HUMAN REVIEW** — the candidate cleared every gate. If any gate fails (try **strict gates** in the sidebar), the banner reflects the hold and the case routes through an escalation step before the doctor signs off.
- **Autonomous engineering record.** A terminal card showing the commit the engineer agent prepared for its bounded pipeline-contract repair. With **SHERLOCKML_AUTOCOMMIT=1** it makes a real local commit and the card shows the SHA; by default it stays in safe “proposed” mode.
- **Case artifacts.** Chips linking the generated Markdown report, the machine-readable JSON report, and the pipeline repair diff — every claim on screen has a file behind it.

What to say: “This is a genuinely retrained model evaluated on the same incident data — F1 back from 0.49 to 0.79, every gate passed, the diff and the report saved. And the final act is asking a human for permission. That’s autonomy you can actually ship.”

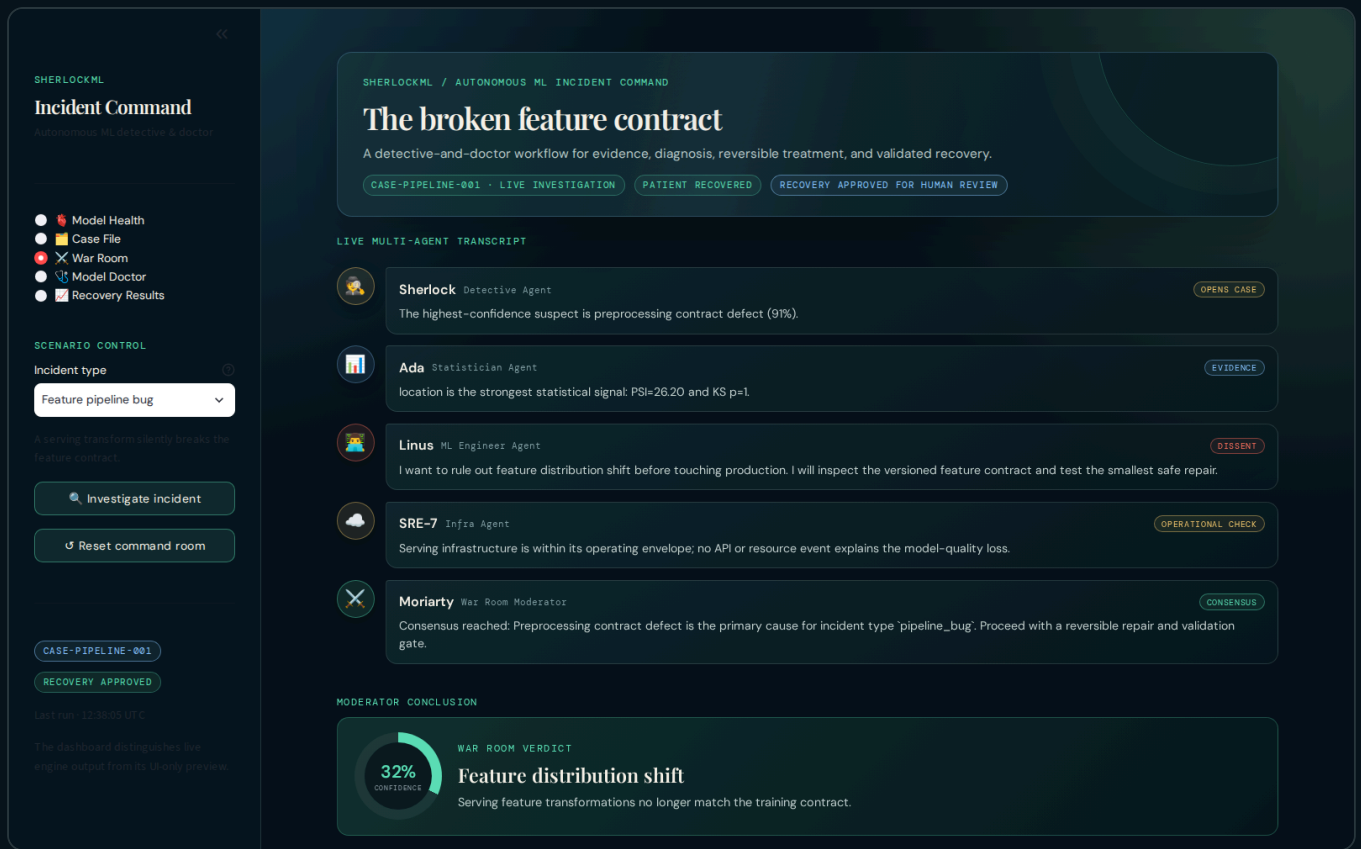
7. The other two scenarios

The same workflow diagnoses two more failure modes. Each produces different evidence, a different debate, a different repair — and a different chart fingerprint.

Feature pipeline bug — CASE-PIPELINE-001

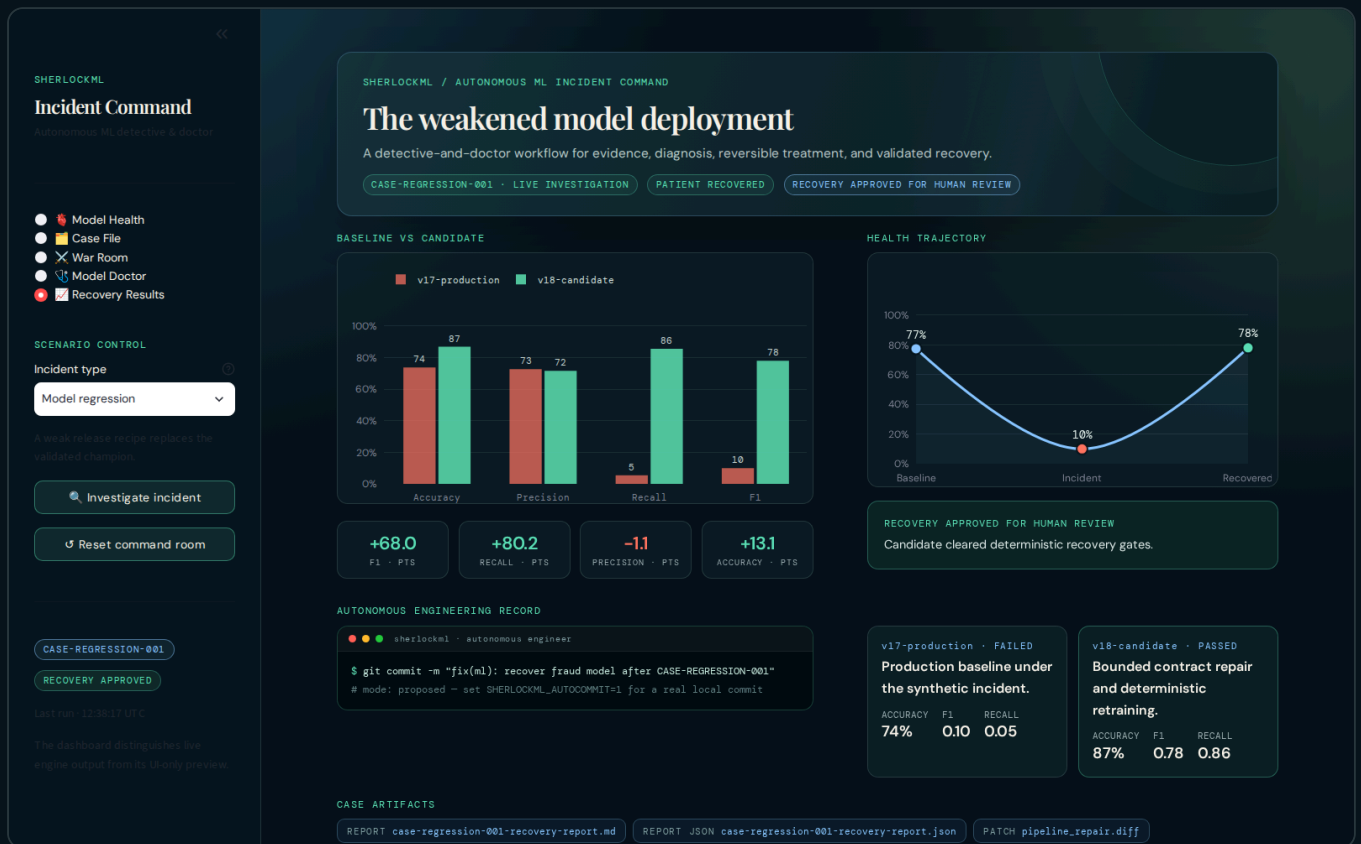


A serving release silently rescales numeric features and breaks category mappings while the *world stays the same*. The drift fingerprint looks completely different: categorical features (`location` , `merchant_category`) spike to extreme PSI because production categories no longer match training, and the suspect ledger flips — *Preprocessing contract defect* becomes primary at 91%.



The War Room reaches a different consensus, citing the contract evidence, and the repair restores the validated preprocessor instead of retraining on new data.

Model regression — CASE-REGRESSION-001



A weak two-feature model was deployed over the validated champion; the data itself is stable. Health collapses to 10%, and the recovery — rolling back to the champion recipe — is the most dramatic: recall from 0.05 to 0.86 and F1 +68 points, all visible in the comparison chart.

Reset between runs with  **Reset command room**; it restores the deterministic pipeline-contract baseline so every demo starts from the same state.

8. What is real, and what is bounded

Honesty is part of the pitch:

- **Real:** the datasets (synthetic but generated fresh each run), the trained XGBoost / logistic-regression models, the PSI and KS statistics, contract-driven retraining from `models/pipeline_contract.json`, the validation gates, the diff artifact, the MLflow (SQLite) experiment record, and the generated reports.
- **Bounded by design:** no customer data, no hosted LLM required, no remote pushes, and no automatic deployment — the strongest outcome the system can produce is *approved for human review*.
- **Deterministic:** a fixed seed makes every scenario repeatable, which is exactly what you want mid-demo.

9. 90-second talk track

1. **Model Health** — “This model didn’t crash. It got sick. F1 fell from 77% to 49% while it kept answering requests.”
2. **Investigate** — “One click. A detective, a statistician, an SRE, an engineer and a doctor work the case in seconds.”
3. **Case File** — “Evidence first: three exhibits, three suspects, and the drift chart showing `transaction_amount` 24× over the threshold.”
4. **War Room** — “They argue. The engineer dissents — then cross-examines the contract. The moderator resolves the case only after the dissent is answered.”
5. **Model Doctor** — “Diagnosis, severity, a three-step prescription, and a recovery estimate derived from the gate results.”
6. **Recovery Results** — “The prescription was tested: a real retrained model, F1 back to 0.79, every gate passed, the fix committed — and the last act is asking a human to approve. Detective, doctor, recovery.”

SherlockML is a deterministic, offline-first technical demonstration. All data is synthetic; recovery approval is a recommendation for human review, never a deployment.